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ORIGINAL RESEARCH

Stochastic assessment of frequency support from wind power plants for power system with high wind penetration using correlation between wind farms

Yeuntae Yoo[1] | Gilsoo Jang[2] | Seungmin Jung[3]

[1]Department of Electrical Engineering, Myongji University, Yongin-si, Korea

[2]Department of Electrical Engineering, Korea University, Seoul, Korea

[3]Department of Electrical Engineering, Hanbat National University, Daejeon, Korea

Correspondence
Seungmin Jung, Department of Electrical Engineering, Hanbat National University, Daejeon, Korea.
Email: seungminj@hanbat.ac.kr

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Abstract
The increase of renewable energy use in a power generation decreases the flexibility of power supply. The intermittent and variable nature of renewable resources necessitates the use of a transmission system operator to prepare reserve generating power in case of power imbalance. Unfortunately, the generation and reserve margins of renewable energy generators are often insufficient. To ensure a balanced and resilient power system, surplus power generation from renewable resources must provide frequency support with the essential number of synchronous generators being remained in the network. Assessment of the amount of frequency support power from wind farm became available through forecast and real-time operation planning. However, the exact amount of back-up power that expected to be provided by individual wind farm can be barely determined due to the uncertainty and variation of renewable resources generation. A stochastic method for assessing the required frequency support operation of specific wind power plants is thus proposed in this paper. The stochastic correlation characteristics between a single wind farm and power system stability are applied in this method. The essential metrics for deloading operation and inertia response are calculated using the stochastic correlation model based on copula function.

1 | INTRODUCTION

The high penetration of renewable energy has induced increasing utilisation of renewable energy generators in power generation recently with conventional electric generators being largely replaced. Unfortunately, the intermittent and stochastic nature of renewable resources causes the lack of availability of controllable generating power [1], making it challenging to secure reliable power system with high penetration renewable energy generators [2]. In contingency situations such as transmission line or generator trip events, the role of synchronous generators is essential as they were the major source of frequency control. Thus, many applications are introduced to compensate for the absence of conventional generators on demand balancing and maintaining stability.

Often referred to as synthetic inertia and droop response, some of the renewable generators can follow the frequency of the power system to mitigate a deviation of frequency from the normal operating range. The power converter that capable of four-quadrant operation enables the fast response from interconnected renewable resources enough to follow frequency deviation as the synchronous generator does through their electro-magnetic coupling. State-of-the-art photovoltaic (PV) generators or wind turbines can be utilised as synthetic inertia for the power system as they have a power converting system that covers their capacity. Unlike fossil-fuelled generators, wind and PV have no inherent energy as their generations are dependent on natural phenomena such as wind speed and solar irradiation. Thus, synthetic frequency support from renewable resources is limited and hard to be quantified.

Considering that renewable energy generators with MPPT produce relatively high amount of power, they can contribute to the supply-demand balance in power systems when operated

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in curtailment mode. This is done through the de-loaded operation in which the power output of renewable resources is accurately forecasted and the required reserve is assessed. Wang et al. [3] presented a statistical approach that investigates the effect of intra-hour wind power variability on frequency response that can be provided by wind turbines. Wind turbines were found to be able to provide an inertia response that is similar to those of synchronous generators using the rotational mass of blade. Besides that, the implementation of additional control loop in the power electronic converter of wind turbine was also proposed in ref. [4] to synthesise inertia. Such artificial inertia, also known as synthetic inertia, was reported to enhance the frequency stability of the distribution network. Despite the findings, the inertia response from wind turbines does not last more than 5 to 10 s [5]. The efficiency of wind power production was also revealed to be relatively low upon the provision of inertia response from wind turbines due to the deviation of the operating point from the optimal condition.

The frequency support from renewable energy generators to the grid thus becomes essential, especially when the renewable energy penetration increases [6, 7]. A tremendous amount of research has been devoted to optimising the frequency support response. In this regard, several techniques for harnessing energy stored in the rotating mass of wind turbine properly were particularly proposed [8, 9]. The amount of frequency support reserve requirement in the system was also assessed depending on the weather condition as demonstrated in refs. [1, 10], and [11]. A proper amount of reserve can be secured by deloading the renewable energy generator or supplying spinning reserve at proper timing.

The frequency support response from a wind turbine can be optimised through various schemes, yet the quantity of support depends on the status of wind speed intake to the turbine. Thus, quantifying the frequency support capability of the wind turbine is tackled by the uncertainty of wind variation. A number of researches have been conducted on scheduling and operating power systems considering the frequency support capability of the wind turbine. In ref. [3], the author assesses the impact of wind power variability on the volume of primary reserve in the power system using statistic data. For operation scheduling of power system with synthetic inertia from wind turbine, a frequency constrain can be included in solving unit commitment (UC) optimisation problem [12].

In the planning stage of the power system, however, it is hard to take frequency support from a wind turbine into consideration for several reasons. Firstly, it cannot be quantified in fixed value. For instance, the volume of synthetic inertia from wind turbine keeps changing and aggregated support capabilities from multiple wind turbines is mixed into another uncertainty distribution. The status of wind turbine operation has a high correlation with other wind turbines that are located adjacently. Thus, the aggregated support capabilities from multiple wind turbines have joint probability distributions with marginal distributions of each turbine. Also, the parameters of controller hardware of renewable energy generators are difficult to be modified in real-time or remotely. Furthermore, a sudden change of operating point of wind turbine could evoke substantial mechanical stress to the turbine itself. The inertia response from the wind turbine has to be considered only for the contingency situation and not for frequency balancing for daily operation. Therefore, it is essential to pre-define the frequency support parameters in the planning stage considering the adequate volume of aggregated frequency support capabilities from multiple wind turbines with correlation characteristics.

There have been very few research efforts devoted to quantifying the amount of frequency support capabilities from wind turbines, especially for inertia response. In ref. [13], considered the constant synthetic inertia support by wind turbine for unit commitment study. In ref. [14], the author conducted frequency constrained planning considering inertia response from a wind turbine. A pre-defined percentage curtailment curve is used for assigning synthetic inertia of wind turbines, yet did not fully illustrate the stochastic dependence between multiple wind turbines and aggregated support capabilities.

The relationship between individual turbine and system stability can be derived from the study on the correlation between wind turbines in adjacent area of wind farms since the wind is the air movement. The modelling of stochastic dependence in uncertainty analysis of renewable energy generator was presented in ref. [15]. Copula function and Sklar's theorem provide means to accurately model joint distribution of random variables such as wind speed or photovoltaic (PV) irradiance at certain regions. Apart from these, a method for modelling conditional forecast errors for multiple wind farms using correlation between random distributions was reported in ref. [16]. The geographical correlation of wind power generation in the European continent was also analysed for wide-area interconnection planning [17].

In this paper, the spatio-temporal correlation between wind turbines is analysed for power system planning to assess frequency support capabilities of wind turbines in terms of their stochastic contribution to the level of wind penetration. The stochastic correlation between the uncertainty factors is built based on the copula theory. The uncertainty of multiple wind turbines is jointly coupled as rank correlation by transforming them into uniform distribution using its own distribution, thus no information is lost during the transformation. The covariance matrix is calculated between distributions of wind turbine generation and n-dimension multivariate probability density function (PDF) is built. The system inertia is assessed in terms of wind turbine penetration level in the system as a form of a conditional probability density function. By combining two marginal distributions that built as a form of conditional PDF, a non-linear curve can be fitted as an inertia constant for each wind turbine that changes based on their contribution to the aggregated inertia support capabilities and status of an intake wind speed. A case study is conducted based on the field measure value of wind speed and actual load demand data gathered from Jeju island in Korea. The contribution of the paper are as follows:

- Proposed a method to calculate system-wide inertia support capabilities considering stochastic correlation between multiple wind turbines in planning stage.

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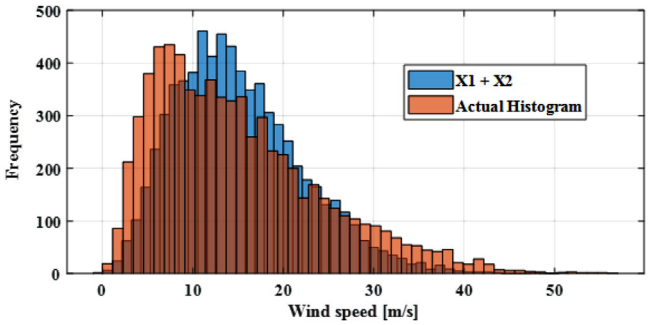

FIGURE 1 Different joint distribution of two random variables by dependency

- Two-stage conditional probability distribution function is introduced to define statistical correlation between the output of wind generators and the status of system inertia.
- Assess the inertia support requirement for the individual turbine considering their stochastic contribution to the aggregated inertia support capabilities of the entire network.

This paper is organised as follows: In Section 2, the concept of joint probability distribution is proposed to correlate stochastic indices in the network. The joint probability distribution between correlated wind speed distributions is further analysed. In Section 3, the probabilistic relation between wind penetration and the system reliability in the network is built based on '*t*' copula function. In Section 4, the proposed method is implemented in the actual power system with high wind penetration level to validate the practicality of the method.

2 | STOCHASTIC CORRELATION IN THE HIGH WIND POWER SYSTEM

Considering that the conventional approach requires extensive scenario analysis, a number of probability density functions are normally combined into a single function such as the aggregated wind power generation in certain area.

One of the approaches is Monte-Carlo simulation. For example, consider two random variables as wind speed from two individual wind farms site A and B. The probability distribution of wind speed at each wind farm can then be developed from historical operation data of each wind farm.

Let x_1 and x_2 be the random variables of wind speed data measured at two offshore wind farms. The distribution of variables X_1 and X_2 can be described as Weibull distribution [18]. When building Monte-Carlo scenarios for summation of these two distributions, random scenarios are created based on the probability density distribution of each variable. Adding $X_1$and X_2 in random order creates another probability density distribution as shown in Figure 1 as blue bar histogram.

However, actual wind speed distributions are geographically correlated which can lead to different forms of probability distribution. The actual distribution of $X_1 + X_2$ can be found in Figure 1. When distributions of each variable are weakly correlated or independent, joint distribution tends to have larger probability near mean point and smaller probability at extreme points. Also, distributions of meteorological parameters are strongly correlated when they are closely located and vice versa. Therefore, a simple summation of two random distributions using Monte-Carlo simulation does not lead to an accurate probability scenario. The joint probability distribution of two random variables has to be modelled based on their correlation.

The correlation coefficient of two random variables can be a measure of dependence between them. This is known as the Pearson correlation coefficient which can be calculated from:

$$\rho(x_1, x_2) = \frac{1}{N-1}\sum_{i=1}^{N}\frac{x_{1,i}-\mu_{x_1}}{\sigma_{x_1}}\frac{x_{2,i}-\mu_{x_2}}{\sigma_{x_2}} \tag{1}$$

where N is the number of variables, μ_{x_i} and ρ_{x_i} are mean and standard variable of each distribution, respectively. However, the correlation coefficient cannot provide information on how joint distribution is formed. Moreover, such coefficient depends on the actual value of the marginal distribution, suggesting that it is affected by the transformation of the marginal distribution. Assume that the wind turbine output is proportional to three times of wind speed in two areas. In this case, the correlation coefficient between wind speeds may differ from correlation coefficient between wind power generated. This drawback induces the development of the concept of rank correlation and copula theory.

Rank correlation ρ_r between two random variables X_1 and X_2 can be defined as:

$$\rho_r(x_1, x_2) = \rho(F_{x_1}(x_1), F_{x_2}(x_2)) \tag{2}$$

where F_{x_1} and F_{x_2} is the cumulative distribution function of each random variables. When applying own cumulative distribution function to a random variable, it transformed to rank parameters between 0 and 1. And the distribution of variables turns into a uniform distribution.

The transformation of the random variable X_1 into a uniform domain using a cumulative distribution function is demonstrated in Figure 2. The cumulative distribution function is driven empirically by data of original distributions. In the uniform domain, the correlation of two random variables can be measured by rank. Since rank correlation tends to have more acceptance against non-linear correlations, a joint probability distribution by two random variables can be modelled more linearly.

A copula is a multivariate cumulative distribution function for a uniform marginal probability distribution. One can state multivariate joint distributions in terms of univariate marginal distributions by using Sklar's theorem, [19]. In this regard, the bivariate distribution with density h can be expressed in terms of marginal distributions as follows:

$$h(x_1, x_2) = c(F_{x_1}(x_1), F_{x_2}(x_2)) \tag{3}$$

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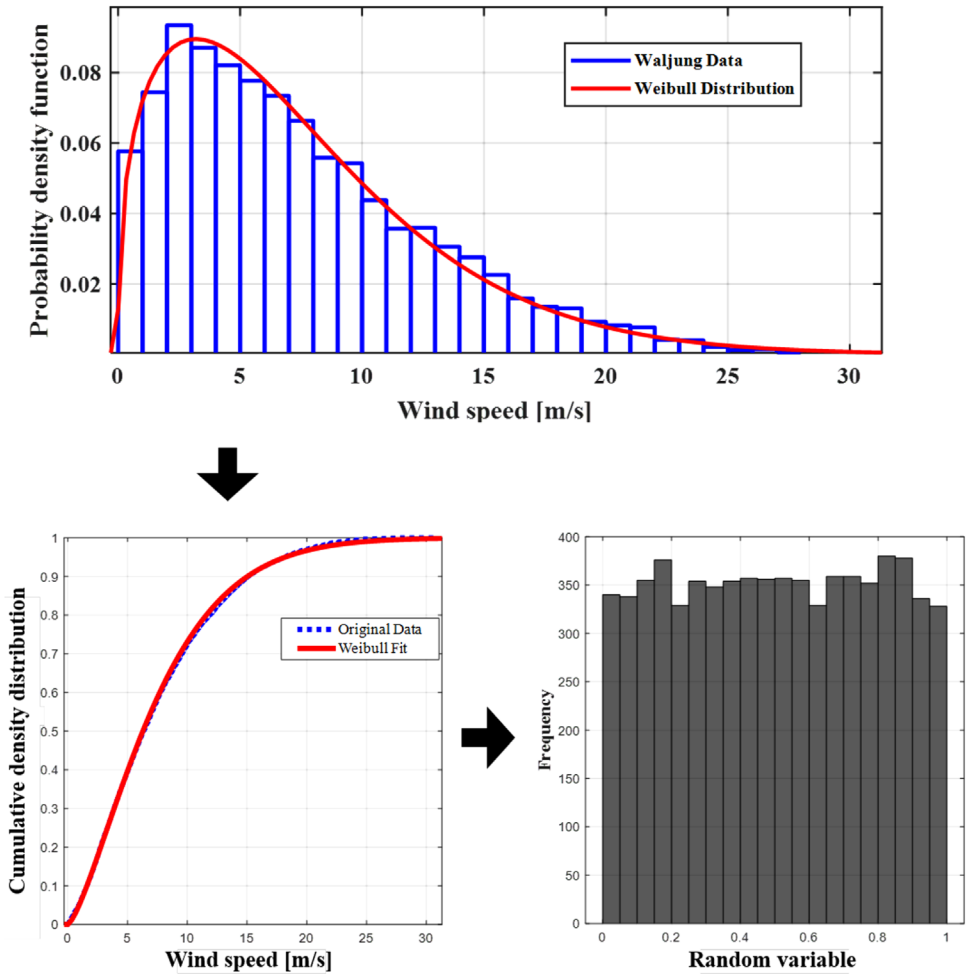

FIGURE 2 Transformation of random distribution to the cumulative distribution

where c is the density of the specific copula. To get copula function as a joint probability distribution, several types of fitted function of copula can be considered.

3 | BUILDING CORRELATION BETWEEN STOCHASTIC FACTORS IN HIGH WIND POWER SYSTEM

When planning to build wind power plants in certain area, the quality of wind resources must be evaluated in advance as it considerably affects the average wind power generation. A wind farm with high utilisation ratio has higher possibility to be exposed to high-speed wind compare to that with low utilisation ratio. For this reason, wind turbines are often concentrated in a specific area although they might be built by different suppliers. The high geographical adjacency results in high correlation between the wind turbines on their generation. When a wind turbine receives fast wind and generates maximum power, other wind turbines likely operate in rated power condition. The wind turbine or farm that contributes to higher level of aggregated wind generation throughout the network can be determined via the correlation between wind power generators in the system. Such a correlation can also be used to assess the required frequency support response from each wind turbine in terms of intake wind speed. This is because the change of system reserve requirement is highly dependent on the capacity of renewable generations at a specific time. As shown in Figure 3, two-stage rank correlation is built to define relation between three parameters, which are the largest capacity of possible $N-1$ contingency, the total wind power penetration level, and the available power of each wind turbine. When the capacity of wind farms are larger than synchronous generators in the system, one of the wind farms can be the plant that generates the largest power in the system. The scenarios will possibly be observed where a system has large-scale renewable generators with relatively small load demand.

3.1 | Stochastic correlation using fitted copula

Assuming the joint probability distribution between the single wind farm and the total wind generation in the system is

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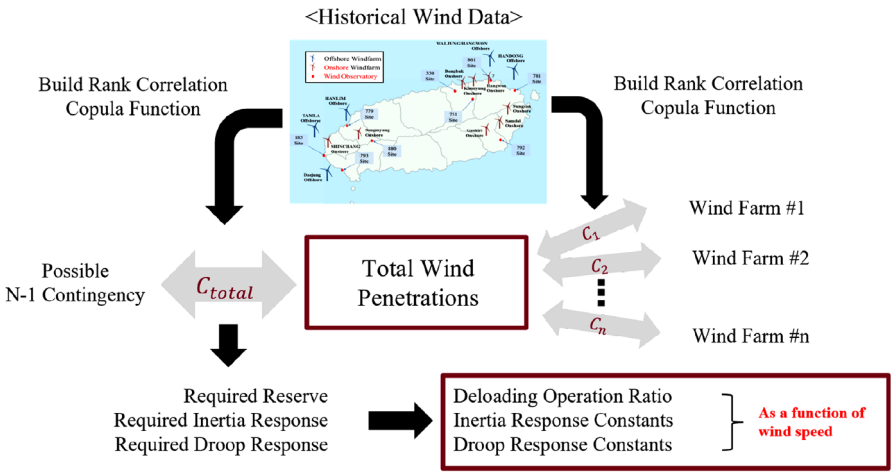

FIGURE 3 Stochastic correlation between multiple wind turbines and system-wide inertia for frequency support capability assessment

expressed as follows:

$$c_{\text{total}}(x_i, x_{\text{total}}) = c(F_{x_i}(x_i), F_{x_{\text{total}}}(x_{\text{total}})) \tag{4}$$

where x_i is a random variable of output from ith turbine in the entire power system while x_{total} is the random variable of the entire production of all wind turbines. Both distributions can be empirically assessed with adequate amount of data sets. F_{x_i} and $F_{x_{\text{total}}}$ are the cumulative density distribution of each random variable. The cumulative density distribution can also be empirically assessed.

The amount of reserve requirement in the power system is closely related to the capacity of the largest generator within the network. The worst $N-1$ contingency scenarios can happen during the failure of a large wind farm. If the correlation between reserve requirements for $N-1$ contingency scenario and total wind generation output exists, it can be expressed in:

$$c_{N-1}(x_{N-1}, x_{\text{total}}) = c(F_{x_{N-1}}(x_{N-1}), F_{x_{\text{total}}}(x_{\text{total}})) \tag{5}$$

where x_{N-1} is the distribution of the maximum online generator's capacity. The required reserve capacity can be assessed depending on the total renewable generations in power system by using ref. (5). Assume that inverse of F_{x_n} and $F_{x_{\text{total}}}$ exist and when the possible output of $x_{\text{total}} = A$, there exists a highest possible $x_{N-1} = B$ which can be defined as:

$$c_{N-1}(B, A) = \max(c_{N-1}(x_{N-1}, A)) \tag{6}$$

In this case, the required reserve capacity B becomes a representative $N-1$ contingency case when total system renewable output is A. As shown in Equation (7), deloading ratio r_{dld} of total renewable generation can be simply defined as B/A, which changes depending on the total output of renewable generation and the corresponding required reserve.

$$r_{\text{dld}}(x_{\text{total}}) = \frac{F^{-1}_{x_{\text{total}}}(c_{N-1}(B, x_{\text{total}}))}{x_{\text{total}}} \tag{7}$$

From Equation (4), the relationship between x_i and x_{total} can be defined as:

$$x_{\text{total}} = F^{-1}_{x_{\text{total}}}(F_{x_i}(x_i)) \tag{8}$$

Deloading operation ratio that corresponds to the current output of generator can be derived from the combination of Equations (7) and (8), which becomes:

$$r_{\text{dld}}(x_i) = \frac{F^{-1}_{x_{\text{total}}} max(c_{N-1}(B, F^{-1}_{x_{total}}(F_{x_i}(x_i))))}{F^{-1}_{x_{total}}(F_{x_i}(x_i))} \tag{9}$$

The deloading operation curve for every renewable generator in the power system can be obtained using (9). This equation considers the correlations between each variable which results in different deloading operation curves among renewable generators. If reserve from the remaining synchronous generators P_{res} is considered, the required deloading operation ratio can be rewritten as:

$$r_{dld}(x_i) = \frac{F^{-1}_{x_{total}} \max(c_{N-1}(B, F^{-1}_{x_{\text{total}}}(F_{x_i}(x_i)))) - P_{\text{res}}}{F^{-1}_{x_{\text{total}}}(F_{x_i}(x_i))} \tag{10}$$

3.2 | Frequency reliability parameters for reserve

The frequency support from a wind turbine can be categorised into inertia and droop response as shown in Figure 4. Similar to the synchronous generator, wind turbines have large kinetic energy in the rotating blade. With an additional controller, this energy can be used to emulate inertia for frequency support. This process temporarily makes the wind turbine's operation deviate from the maximum power tracking condition. The decrease in efficiency is critical for some contingency scenarios in the grid and thus, the inertia response of wind turbines should be carefully assessed. In case of droop response, the

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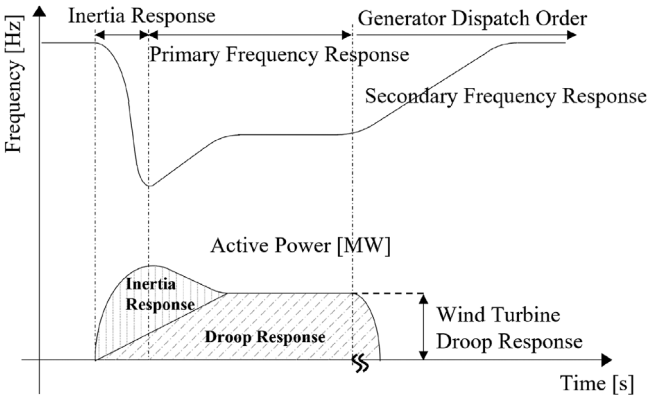

FIGURE 4 Primary frequency support from wind turbine

wind turbine should be in a deloading mode. The deloading is a concept of shifting the operation of a variable speed wind turbine(VSWT) below its maximum power point (MPP). A wind turbine in deloading mode can provide the increased generation output for a longer duration as shown in Figure 4.

Assume the numerator term of Equation (10) as $\triangle P_{\max}$ and the denominator term as S_{wind} when determining frequency response parameters of the wind turbine. Maintaining (rate of change of frequency) ROCOF within ±0.5 Hz/s was recommended in ref. [20] to ensure the reliable operation of the power system. The above ROCOF standard was used as a reference herein to validate the proposed algorithm. The minimum inertia response parameter of wind turbines can be evaluated using the swing equation and the maximum limit of the ROCOF value, which is as follows:

$$H_{\text{wind}} S_{\text{wind}} \geq \frac{\Delta P_{\max} f_0}{2\frac{\mathrm{d}f}{\mathrm{d}t}} - \sum_{i=1}^{N} H_i S_i \tag{11}$$

where N is the number of the conventional generators; S_i and S_{wind} are the rated capacity of individual generators and wind turbines, respectively, and H_i and H_{wind} are the inertia constant of each generator and wind turbines, respectively.

The primary frequency response support of the wind turbine needs to be assessed along with the inertia response to satisfy the steady-state operation constraint of the given network. The most important security constraint related to frequency is the under-frequency load shedding (UFLS). The minimum required droop constant can be calculated by assuming a minimum frequency limit as the UFLS operating frequency.

$$\frac{1}{R_{\text{wind}}} \frac{S_{\text{wind}}}{f_0} \geq \frac{\delta P_{\max}}{f_0 - f_{\min}} - \sum_{i=1}^{N} \frac{1}{R_i} \frac{S_i}{f_0} \tag{12}$$

where $f_{\min}$ is the minimum frequency and R_i is the droop constant of individual generator. The contribution of frequency response from wind turbine can be implemented with the simple control loop [21]. Assume that a wind turbine has additional active power control loop as shown in Figure 5. Then, the

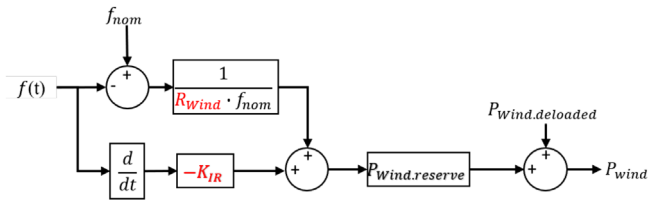

FIGURE 5 Wind turbine controller for frequency support

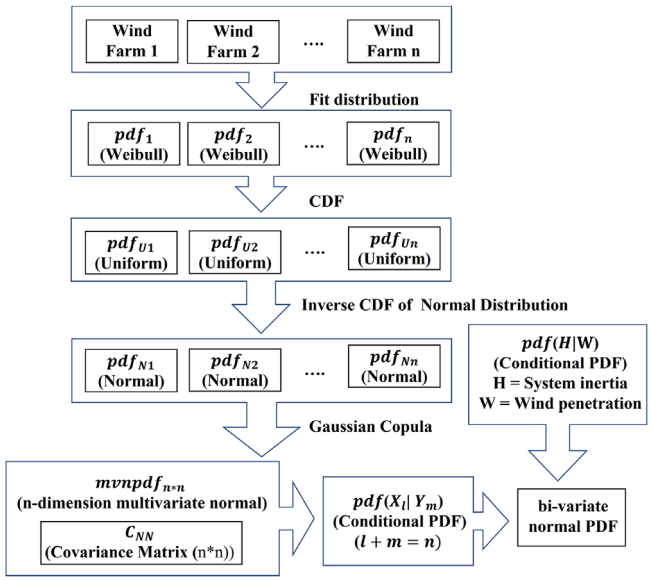

FIGURE 6 Process of stochastic correlation analysis between multiple wind turbines for frequency support capability assessment

frequency response of the system receives additional droop response from the wind turbine with droop constant R_{wind}. The inertia time constant H_{wind} of the wind turbine with the given control loop can be defined as:

$$H_{\text{wind}} = K_{IR} \frac{f_{\text{nom}}}{2} \tag{13}$$

where f_{nom} is the nominal frequency of the network and K_{IR} is the inertia coefficient in the control loop.

From Equations (11) and (12), one can calculate inertia and droop constant of an entire fleet of wind turbines against the most probable '$N-1$' contingency case. The relationship between each wind farm and total wind turbine output can be known from Equation (5). This consecutive relationship results in non-linear characteristic curves that relate the frequency response parameters to the current wind speed input. The overall process can be found in Figure 6. As the wind generators have no inherent inertia, the status of system wind generations has no direct influence on system-wide inertia. The system-wide inertia is determined mostly by the number of synchronous generators and synchronous condensers that remain in the network. As the output of wind generators increases, those sources of natural inertia are displaced, and the inertia of the entire system decreases. Since there is a minimum requirement on the

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number of synchronous generators that must remain in the network, surplus wind generations have to be curtailed. The system-wide inertia can be calculated considering the number of synchronous generators and their rated capacity using Equation (11). The inertia of wind generators differs depending on their size and weight. The evaluation of the inertia of the wind generator system should be carried out extensively in the operation stage of the power system. The purpose of this study is to assess the mandatory capacity of inertia response from each wind turbine depending on their capacity factor and contribution to $N-1$ contingency capacity. In planning stage of power system, it is hard to assess the quantity of inertia response when wind turbines in the system are assumed to have synthetic inertia. In this study, the capacity factor of each wind turbine is combined into a form of joint probability density function considering the correlation of generation among them. Through this, the inertia is assigned to each wind turbine and makes the wind turbine can synthetically form the mandatory inertia with a modified inverter controller.

As shown in Figure 6, Gaussian and student's 't' copula is used to build stochastic correlation. The correlation between wind turbine generators can be defined as a co-variance matrix. Elliptical copulas, such as Gaussian and student's 't' copulas, are known to have advantages over other families of copulas in terms of specifying different levels of correlation between marginals. Since elliptic copulas are more feasible for defining elliptically contoured distributions, wind speed distribution is used instead of the using a distribution of wind generation. Distributions of wind generation have a substantial amount of data points in the tails of distribution while the distribution of wind speed has most of its data near the median. Thus, joint probability density among wind speed distributions has elliptical shape and for this reason, Gaussian copula is used. On the other hand, bivariate distribution of aggregated wind generation and maximum $N-1$ contingency capacity has more points in the tails, and therefore the student's 't' copula is better suited to describe the distribution.

4 | CASE STUDY IN REAL POWER SYSTEM

4.1 | Introduction to the studied wind power system

To verify the practicality of the relationship between the wind speed correlation and system reliability index, the field measured data of Jeju island at the Korean peninsula is used. Jeju island has relatively high wind energy penetration level in the peninsula. Table 1 presents the detailed parameters of the facilities in the Jeju island power system. As the penetration level of renewable generators increased, the displacement of synchronous generators continued until three must-run generators remained in the network, which was a crucial operating constraint to sustain the voltage and frequency of the power system. The assumption that no other generating sources are dispatched during the simulation was also made.

TABLE 1 The configuration of the studied wind power system in Jeju island

	Rated capacity [MW]	Droop constant	Governor type	Inertia constant [s]
GEN 1	80	0.05	IEEEG1	6
GEN 2	80	0.05	IEEEG1	3.735
GEN 3	45	0.05	GAST2A	5.941
Wind farm	696.56	Mostly type 3 & 4		

Figure 7 shows the geographical distribution of wind farms on Jeju island. Several meteorological stations that measure wind speed near the wind farm are indicated by red dots. The historical data of wind speed are used to build joint probability distribution. 1 year data of wind speed was gathered for simulation. Table 2 shows the correlation coefficient between the measuring sites, and the correlation level among sites is very high. When the geographical distance between sites decreases, the correlation coefficient is found to increase as well. The wind farms in Figure 7 have their corresponding measuring sites. The average wind speed of the corresponding wind farms is calculated using the wind speed data obtained from the measuring sites. Table 3 shows the average wind speed and rated capacity of each wind farm. The corresponding wind speed measuring sites are also listed in the table.

4.2 | Determination of deloading characteristic using correlation

The contribution of specific wind turbine to the aggregated wind power can be determined using the joint probability density function as shown in Figure 8. The turbine in site (a) has lower correlation to the increase of aggregated wind generation compared to a turbine in site (b), which indicates the good wind quality in site (a). The turbine in site (b) has relatively poor output distribution. There is a high chance that power supply in the system will surplus when the wind speed over the site (b) is very high. According to the proposed method, the frequency support response from the turbine in site (a) should higher than that from the turbine in site (b) when high wind intake is expected. On the contrary, when wind speed is low, the turbine in site (a) will provide less frequency support compared to the turbine in site (b).

The correlation between the single wind farm and the total wind power generation needs to be defined in rank correlation to identify the relation characteristic. The cumulative density function of wind farms distribution should be built in advance to develop a rank correlation. To this end, such a cumulative density function is developed using the historical data of wind farm output as shown in Figure 2. Figure 9 displays the transformation of joint distribution to rank correlation between individual wind farm and the total wind farm output in the network. The joint distributions for the rest of wind turbines

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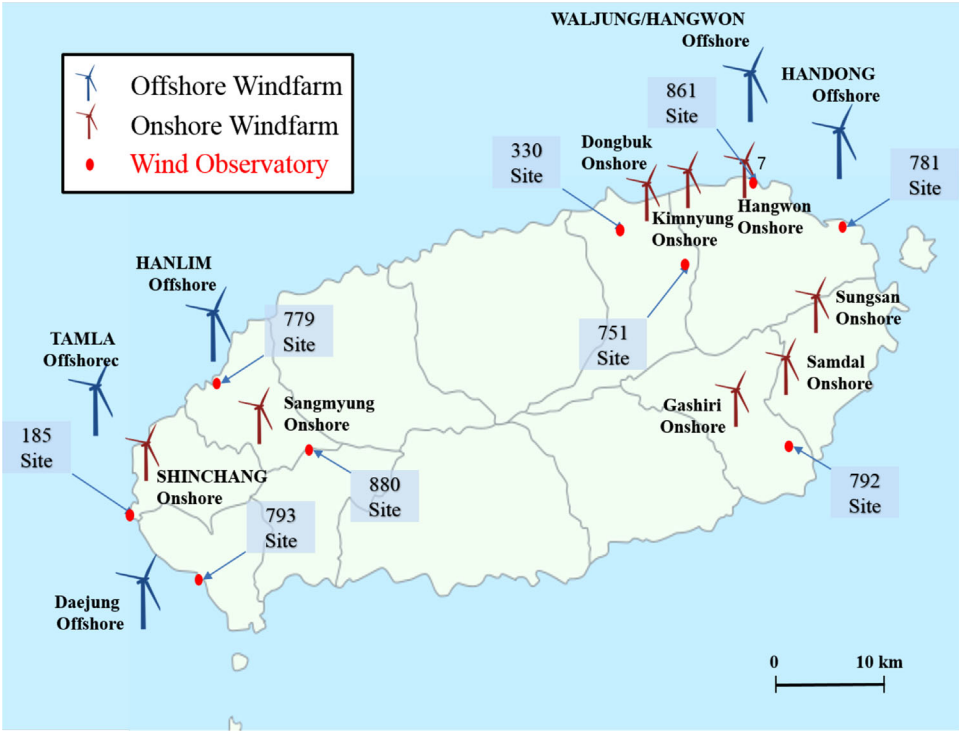

FIGURE 7 The geographic distribution of wind farms and measuring sites of the studied wind power system in Jeju island

TABLE 2 The correlation coefficient between the corresponding measuring sites of wind farms

Site	781	185	330	779	792	793	861	880
781	**1.00**	0.71	0.77	0.69	0.69	0.67	0.84	0.61
185	0.71	**1.00**	0.71	0.65	0.50	0.78	0.74	0.67
330	0.77	0.71	**1.00**	0.62	0.57	0.70	0.76	0.57
779	0.69	0.65	0.62	**1.00**	0.64	0.64	0.63	0.63
792	0.69	0.50	0.57	0.64	**1.00**	0.57	0.52	0.56
793	0.67	0.78	0.70	0.64	0.57	**1.00**	0.62	0.71
861	0.84	0.74	0.76	0.63	0.52	0.62	**1.00**	0.60
880	0.61	0.67	0.57	0.63	0.56	0.71	0.60	**1.00**

TABLE 3 The average wind speed and rated capacity of each wind farm

	Avg. wind [m/s]	Rated capacity [MW]	Grouped to site
Hanlim Off-S	7.29	100	779
TAMLA Off-S	7.48	30	185
Daejung Off-S	7.8	100	793
Waljung Off-S	7.5	125	861
Handong Off-S	7.5	105	781
Gashiri On-S	6.7	45	792
Shinchang On-S	7	22.7	185
Sangmyung On-S	6.5	23	880
Dongbuk On-S	5.7	30	330
Kimnyung On-S	6	41.25	330
Hangwon On-S	7	21.61	891
Sungsan On-S	6	20	781
Samdal On-S	6.5	33	792

have to be developed to relate deloading operation curves and each turbine.

The relationship between the generator output of the entire wind farm and the maximum $N-1$ contingency scenario capacity is shown in Figure 10. The highest possible value curves for a total wind turbine can be fitted from the joint distribution function by Equation (6). The 't' copula that best describes the correlation characteristics is considered for the copula function fitting. The relation between total wind farm output and maximum $N-1$ contingency scenario capacity is assessed in Figure 11. The relation has non-linear characteristics. The trend of the curves changes when total wind farm output distribution changes. The optimal deloading ratio presented in Figure 12 can be calculated using the non-linear relation curve in Figure 11 and Equation (7). When taking synchronous generators into consideration, the calculated deloading ratio of wind turbines can be reduced as described in Equation (10). The red solid line in Figure 12 represents the deloading ratio that assumes the remaining synchronous generators have 50 MW reserves.

Figure 13(a) shows the deloading ratio of each wind turbine determined using Equation (10). The contribution of each turbine is observed to be different according to wind speed since the dependency between each turbine and total wind farm output is different. The inertia and droop constants of each wind

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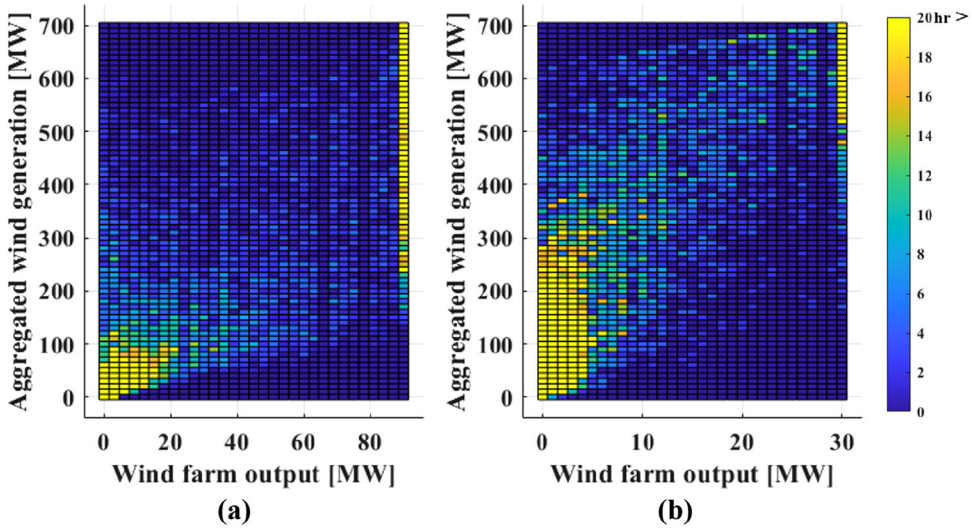

FIGURE 8 The correlation between the single wind farm and the aggregated wind power generation in (a) Daejung Off-S and (b) Dongbuk On-S

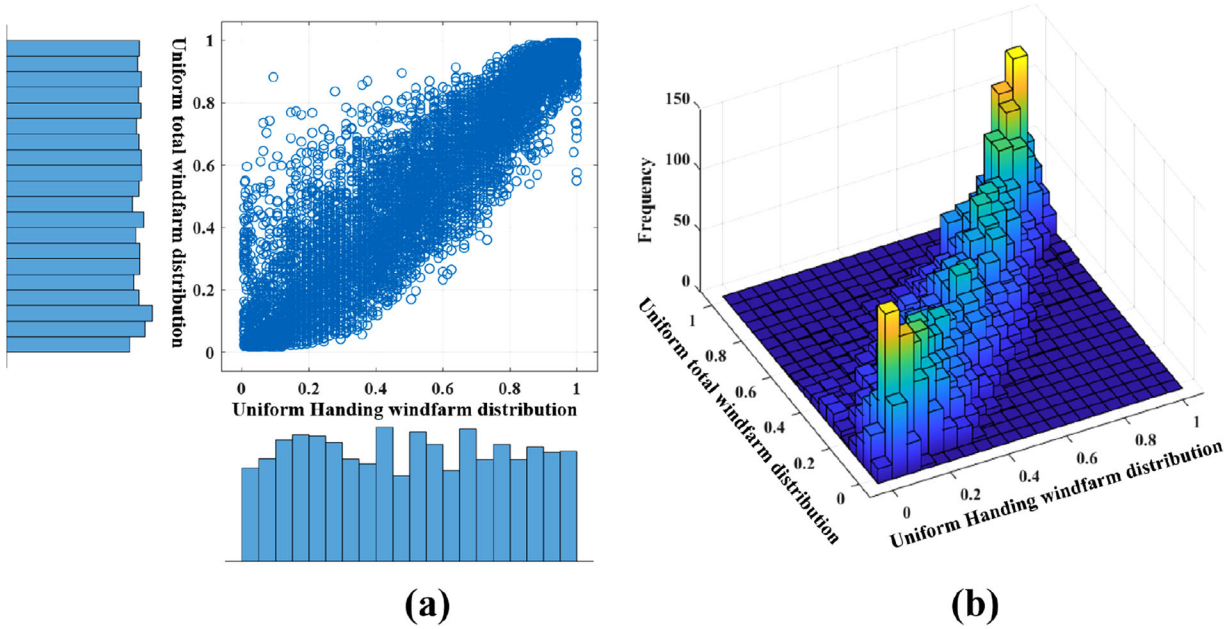

FIGURE 9 The joint probability distribution of Handong wind farm and the total wind farm output (a) rank correlation distribution (b) rank correlation histogram

turbine can be defined using Equations (11) and (12). Assuming the UFLS reference of the system is 0.2 Hz and needs to limit the maximum ROCOF to ±0.5 Hz/s, the inertia and droop constant of each wind turbine can be defined as shown in Figure 13(b,c). The response of each wind turbine has nonlinear characteristics in terms of wind speed. The wind farms are also observed to provide different frequency support response depending on their correlation to the aggregated wind farm generation in the network. The difference in the response constant of Daejung and Dongbuk wind farms is the largest due to the wide disparity in their correlation to the aggregated wind farm generation as depicted in Figure 8.

4.3 | Case study: a year-long wind speed data

The hourly wind speed data of Jeju island was prepared to verify the impact of the proposed deloading operation curve. The wind speed of the entire year of 2018, around 7040 h, was recorded. The hourly average wind speed data measured from each meteorological station was gathered in monotonic order. Due to data losses and error, the number of data is less than the actual time frame of 1 year (7040 h out of 8760 h data). The meteorological data in this paper was collected from the weather station built and operated by the Korean meteorological administration. There are several missing data points in the

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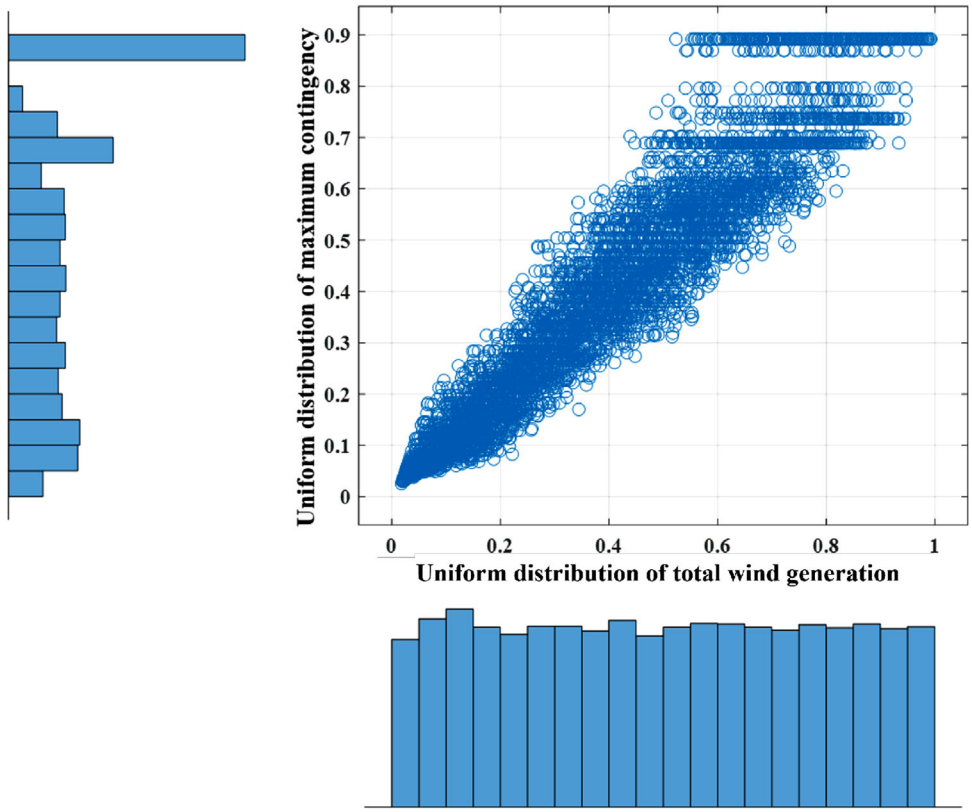

FIGURE 10 The joint probability distribution between total wind power generation and the maximum $N-1$ contingency scenario capacity

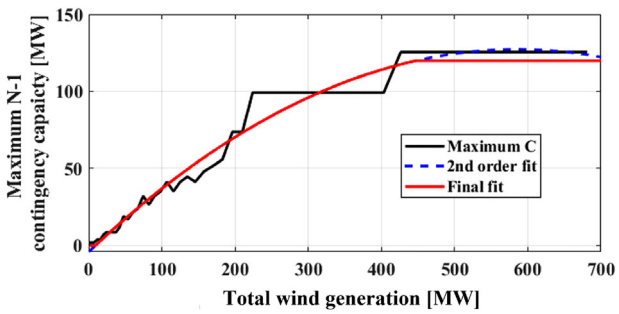

FIGURE 11 The fitted curve of joint probability distribution curve between total wind generation and maximum $N-1$ contingency capacity

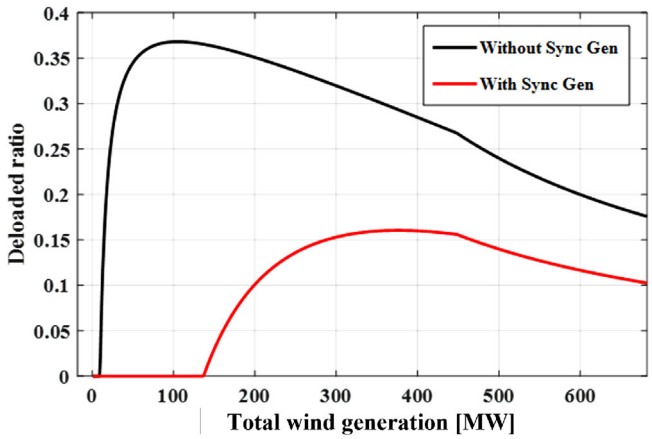

FIGURE 12 The deloading ratio of wind farm in terms of total wind generated

meteorological data set gathered from the weather station. In case study, missing data points and adjacent set are excluded. Once the wind speed data of each wind turbines was collected, curtailment was applied based on the deloading ratio curve shown in Figure 12 and later compared with the method that evenly distributes the deloading ratio to every wind turbine as considered in other literatures [13, 14].

The power system is assumed to operate with the reserve provided from the wind farm curtailment only. For this scenario, the result of a year-long operation can be found in Table 4. The violation hours indicate the cumulative operation hours during which the reserve in power system was insufficient according to Equations (11) and (12). The proposed method has less curtailment amount per year than the other case with the same violation hours. The same procedure can be applied to the case of inertia response by using Equations (10) and (12). The inertia response constant curve in Figure 13 is compared with the method that evenly distributes inertia constant. Since the ROCOF of system is dependent on the net value of cross product of H and S in the system as shown in Equation (10), the net value should be higher enough to prevent the frequency of the system from decreasing rapidly. However, if the net value is too high, the recovery of frequency after contingency might be exacerbated with excessive power being extracted from wind

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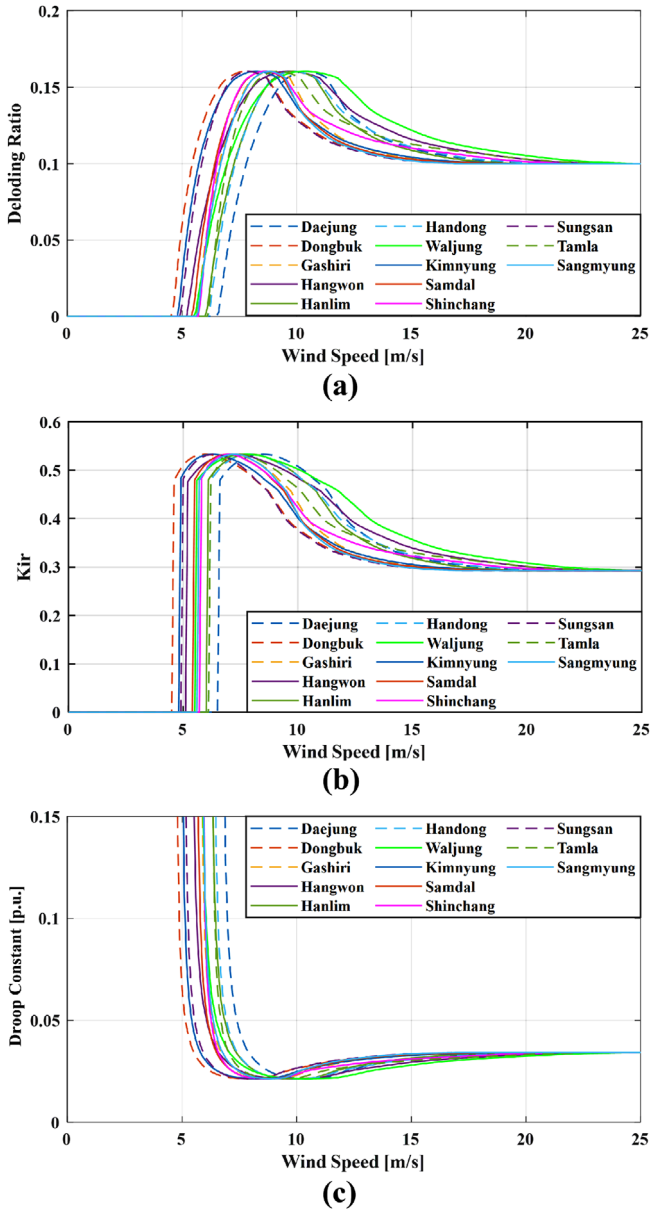

FIGURE 13 The frequency support metrics of wind turbine: (a) deloading ratio, (b) inertia coefficient, and (c) droop constant in terms of wind speed

TABLE 4 The comparison of a year-long operation in terms of curtailed amount and violation hours

	Curtailed amount [MWh] per year	Violation hours [h]
Proposed	**203,806**	**1,957**
5% curtail	83,725	3,554
7% curtail	117,215	3,407
10% curtail	167,451	3,119
12.55% curtail	**210,151**	**1,975**
15% curtail	251,176	1,183
20% curtail	334,902	456
30% curtail	502,353	50

TABLE 5 The comparison of a year-long operation in terms of inertia response amount and violation hours

K_{ir}	Net H·S [s·GW] per year	Violation hour [h]
Proposed	**19,887**	**2,662**
0.2	10,047	4,976
0.3	15,070	4,549
0.396	**19,890**	**2,807**
0.404	**20,320**	**2,665**
0.5	25,117	1,604

power generators. The result in Table 5 shows that the proposed method has less violation hours than the case with same net value of H*S.

When the wind speed intake is high for wind turbines with high correlation to aggregated wind generation, the rest of the wind turbines likely operate with the maximum power. In this case, the frequency support parameters (K_{ir}, droop constant) needs to be set at lower values as there are plenty of wind turbines that can provide back-up power during contingency situation. On the other hand, in case of turbines with low correlation, the frequency support parameters should be higher considering the insufficient back-up power from the rest of turbines.

5 | CONCLUSION

An assessment method for frequency support from wind farms using rank correlation of entire wind farms in the power system is presented in this paper. The correlation characteristics of wind speed data at a different geographical point can be determined using historical data.

The concept of rank correlation is thus introduced. The joint probability density function for every correlated parameter is developed using the copula function. As a result, two probabilistic relationships are established, which are the joint probability density functions between the single wind farm output and the total wind farm output and between total wind farm output and the maximum $N-1$ contingency capacity. The frequency support requirement for individual wind farm in the network can be accurately evaluated by combining these two probabilistic relationships. Each wind turbine has different response characteristics depending on their correlation to the aggregated wind power generation in the network.

A year-long wind speed data in Jeju island is also used to validate the practicality of the proposed method. The empirical probability density function and cumulative density function are developed from the historical wind speed data. The proposed method allows a smaller amount of wind power curtailment compared to the conventional method that adopts a constant deloading scheme. The inertia response requirement can also be efficiently evaluated from the correlation of each wind turbines.

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During the planning stage, the proposed method is very useful for assessing the impact of wind power generator integration into the network since the required inertia response of individual wind turbine is difficult to be assessed using the conventional approach. The introduction of the proposed method can promote the need for storage devices or additional control system for grid support.

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CONFLICT OF INTEREST

The authors declared no conflict of interest.

DATA AVAILABILITY STATEMENT

The grid data that support the findings will be available in KEPCO at [https://www.kepri.re.kr:20808/index] following an embargo from the date of publication to allow for the commercialisation of research findings.

ORCID

Gilsoo Jang https://orcid.org/0000-0001-7590-8345
Seungmin Jung https://orcid.org/0000-0002-9806-9545

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